Example 1.9

Find atleast two rational numbers between $\frac{-3}{4}$ and $\frac{-2}{5}$.

Solution:

The denominators are different for the given rational numbers. The LCM of the denominators 4 and 5 is 20. Make the rational numbers such that they have common denominators as 20. Here,

$\frac{-3}{4}=\frac{-3}{4}\times\frac{5}{5}=\frac{-15}{20}$ and $\frac{-2}{5}=\frac{-2}{5}\times\frac{4}{4}=\frac{-8}{20}$.

It is easy now to find and insert rational numbers between $\frac{-15}{20}$ and $\frac{-8}{20}$ as shown below.

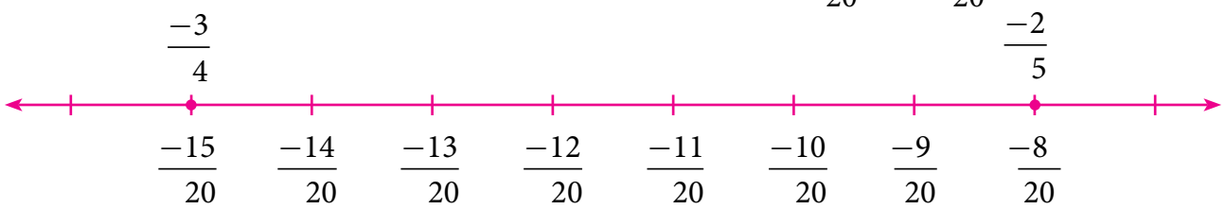


Fig. 1.14

We can list a few rational numbers as $\frac{-14}{20}, \frac{-13}{20}, \frac{-12}{20}, \frac{-11}{20}, \frac{-10}{20}$ and $\frac{-9}{20}$ between $\frac{-15}{20}$ and $\frac{-8}{20}$.

Are these the only rational numbers between $\frac{-15}{20}$ and $\frac{-8}{20}$? Think! Try to find 10 more rational numbers between them, if possible!

Note

We can find many rational numbers between $\frac{-7}{11}$ and $\frac{5}{-9}$ quickly as given below: The range of rational numbers can be got by the cross multiplication of denominators with the numerators after writing the given fractions in standard form. The cross multiplication here $\frac{-7}{11}\times\frac{-5}{9}$ gives the range of rational numbers from –63 to –55 with the denominator 99. This is nothing but making the given rational numbers equivalent with the denominator 99!

Exercise 1.1

1. **Fill in the blanks:**

   (i) $\frac{-19}{5}$ lies between the integers _____ and _____.

   (ii) The decimal form of the rational number $\frac{15}{-4}$ is _____.

   (iii) The rational numbers $\frac{-8}{3}$ and $\frac{8}{3}$ are equidistant from _____.

   (iv) The next rational number in the sequence $\frac{-15}{24}, \frac{20}{-32}, \frac{-25}{40}$ is _____.

   (v) The standard form of $\frac{58}{-78}$ is _____.

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